

Yingrui Hou

Data Scientist / AI Engineer — Industrial ML validation & deployment-ready workflows

Target Application: Safran | AI Engineer F/H

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Introduction personnelle

Ingénieure logicielle scientifique issue de la physique expérimentale, avec plusieurs années d'expérience dans le développement de workflows réutilisables en Python/C++ pour la calibration, la validation et l'inférence sur des systèmes de mesure bruités.

Mon point fort est de transformer des mesures imparfaites et des modèles imparfaits en workflows fiables et exploitables pour la décision, grâce à l'analyse du signal, à l'extraction de paramètres, à la logique de correction et à une validation robuste.

Je suis particulièrement pertinente pour des rôles où la rigueur scientifique, l'automatisation, l'analyse de données expérimentales et la fiabilité de la calibration comptent davantage que l'analytics générique.

Selected Fit For This Role

- Designed and ran reproducible ML pipelines (data prep → training/validation → deployment-ready evaluation) with clear diagnostics.
- Implemented domain-adaptation / reweighting to correct simulation-to-real or dataset mismatch before downstream decisions.
- Built uncertainty/robustness checks and sensitivity analysis to prevent overconfident models in noisy regimes.
- Developed reusable components and shared-code practices in collaborative environments.
- Communicated model limitations and validation results to domain experts, translating into actionable next steps.
- Maintained performance via calibration/correction workflows and before/after comparisons across operating conditions.

the closest match to your core evidence: calibration, signal processing, experimental data, and scientific software.

Education

PhD in Particle Physics
University of Chinese Academy of Sciences
BSc in Applied Physics
China University of Mining and Technology

Languages

Chinese: Native speaker
English: C1
French: B1 to B2

Core Tools & Methods

Tools: pandas | ROOT / RooFit | Geant4 | NumPy | matplotlib / seaborn | Workflow Automation (YAML / Shell / CMake) | Geant4 simulation | statsmodels | Python | C/C++ | etc

Methods: Probabilistic Modelling | Mixture Models | particle-dependent timing models | correlation-based correction | event-level correction | Large-Scale Data Analysis | Detector Simulation | Time-Resolved Signal Analysis | Collaborative R&D | beam-test analysis | etc

Validation: Calibration & Bias Correction | Model Validation | Simulation-to-Data Alignment | Simulation Correction | Simulation Calibration | Performance Diagnostics | Bias Correction | Data Validation | Root-Cause Analysis | Calibration | etc

ML: Domain Adaptation | scikit-learn | CatBoost / Gradient Boosting | Reweighting | Feature Validation | CatBoost training | train-test diagnostics | gradient-boosting reweighting | GBDT | CatBoost | etc

Relevant Experience

Postdoctoral Researcher | 2021 - 2025

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Developed CatBoost-based selection and validation pipelines, including train/test diagnostics, scan studies, and reproducible model application.
- Aligned simulated and observed datasets through gradient-boosting reweighting and control-sample calibration before downstream inference and selection decisions.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.

- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.
- Developed ROOT/C++ calibration and correction workflows to reduce structured bias and compare performance before and after correction across operating conditions.

Selected Project

Domain Adaptation for ML

GBDT | CatBoost | Reweighting | Feature Validation

Built an ML validation and sample-alignment workflow in a shared analysis environment for noisy selection problems under dataset shift.

Methods: CatBoost training, train-test diagnostics, gradient-boosting reweighting

Impact: improved downstream model reliability by treating dataset mismatch and overtraining checks as part of the production workflow.

Industry relevance: Domain Adaptation | Applied ML | Dataset Shift

Test-Data Validation & Root-Cause Analysis

Monitoring | Data Validation | Performance Diagnostics | Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Impact: translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias.

Industry relevance: Production Monitoring | Model Validation | Observability | Quality Assurance

Sensor Simulation & Reconstruction

Signal Processing | Simulation Calibration | Digital Twins

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor...

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction

Impact: linked measured non-uniformity to system-level resolution through both simulation code and experimental validation.

Industry relevance: Sensor Analytics | Measurement Systems | Simulation-Assisted Design